

AI Usage

AI used to build this product

Claude (Anthropic), via the Claude.ai chat interface with the code-execution container, was used as a coding/development assistant for this entire prototype: architecture planning, writing the backend (models, services, controllers, routes, middleware, jobs), the frontend (React components and pages), tests, the evaluation harness. See `docs/DEVELOPMENT_PROMPTS.md` for the actual prompts used, organized by area.

No other coding assistants, debugging tools, or design tools were used.

AI used by the final product

The deployed application itself calls the **Groq API** (`llama-3.3-70b-versatile` by default, configurable via `GROQ_MODEL`) for five distinct features, each logged separately in `AIUsage` for observability:

Feature	Where	What it does
<code>tutor</code>	<code>tutorService.askTutor</code>	Answers learner questions grounded in retrieved Project material; decides (and self-reports) whether evidence is sufficient; may call the <code>getConceptMastery</code> tool.
<code>concept_extraction</code>	<code>materialProcessorService.extractConcepts</code>	Identifies 4-10 key concepts from newly processed material, seeding <code>Concept</code> / <code>Mastery</code> records.
<code>quiz_generation</code>	<code>quizService.generateQuestion</code>	Generates one MCQ or open-ended question grounded in material for a chosen concept + difficulty.
<code>quiz_grading</code>	<code>quizService.gradeOpenAnswer</code>	Grades open-ended quiz answers for understanding/accuracy/relevance, returning explanatory feedback, not just a score.
<code>recommendation</code>	<code>recommendationService.generateRecommendations</code>	Converts mastery evidence into 1-3 specific "what to do next" recommendations.

Retrieval itself (finding relevant material chunks) is **not** an AI call — it's TF-IDF cosine similarity computed in `retrievalService.js` (see `docs/ARCHITECTURE.md` for why). Mastery level updates (`masteryService.applyEvidence`) are also not AI calls — they're a deterministic confidence-weighted formula over quiz results.

Every one of the five AI features above funnels through the single `aiService.chatComplete` wrapper, which logs model, prompt/completion tokens, latency, an estimated cost, and success/failure to the `AIUsage` collection for every call — visible in the Admin Dashboard's AI Usage tab and each Project's Analytics tab.